UNRAMIFIED MORPHISMS OF ALGEBRAIC SPACES ARE THE FINITELY PRESENTED FORMALLY UNRAMIFIED ONES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2828-2838 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2840-2864, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Unramified and formally unramified morphisms of algebraic spaces, etale charts, and etale locality of the properties; the descent is retained.

Source reference. Original lemma lemma-characterize-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2810–2816. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-unramified-omega-zero: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2864–2864. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(1) *The morphism f is unramified,*
(2) *the morphism f is locally of finite type and $\Omega_{X/Y} = 0$, and*
(3) *the morphism f is locally of finite type and formally unramified.*

Proof. Choose a diagram

$$\begin{array}{ccc} U & \xrightarrow{\psi} & V \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

where U and V are schemes and the vertical arrows are étale and surjective. Then we see

$$\begin{aligned} f \text{ unramified} &\Leftrightarrow \psi \text{ unramified} \\ &\Leftrightarrow \psi \text{ locally finite type and } \Omega_{U/V} = 0 \\ &\Leftrightarrow f \text{ locally finite type and } \Omega_{X/Y} = 0 \\ &\Leftrightarrow f \text{ locally finite type and formally unramified} \end{aligned}$$

Here we have used Morphisms, Lemma morphisms-lemma-unramified-omega-zero and Lemma lemma-characterize-formally-unramified. □